Subsystem — Pt1000 Temperature Sensors

Sensor: PTFC102B1G0 — TE Connectivity (PTF family, Class B)
Channels: 8 × Pt1000, analog 8:1 MUX
Files: `include/pt1000.h` · `src/pt1000.cpp`

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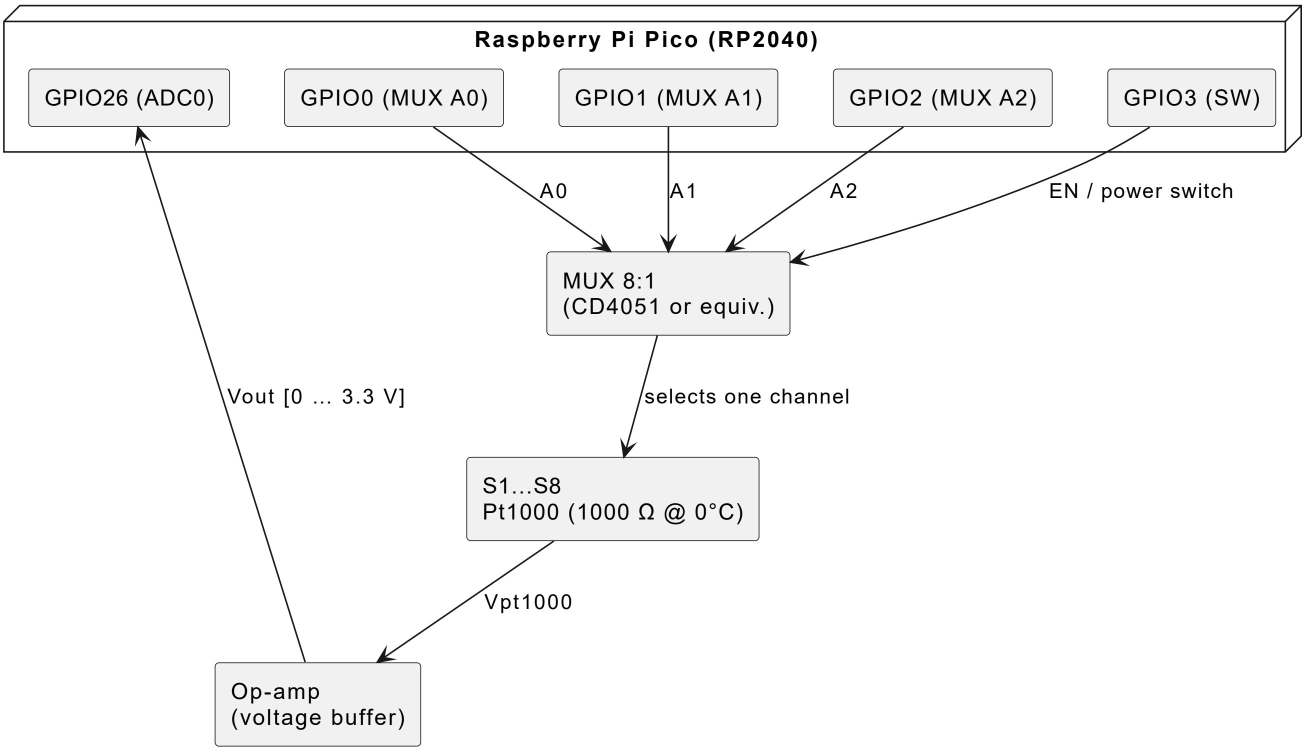
1. Hardware Connections



GPIO	Role	State
GPIO0–2	MUX address A2:A1:A0	OUTPUT
GPIO3	Circuit power switch	HIGH = ON, LOW = OFF
GPIO26	ADC0 — op-amp output	INPUT ANALOG

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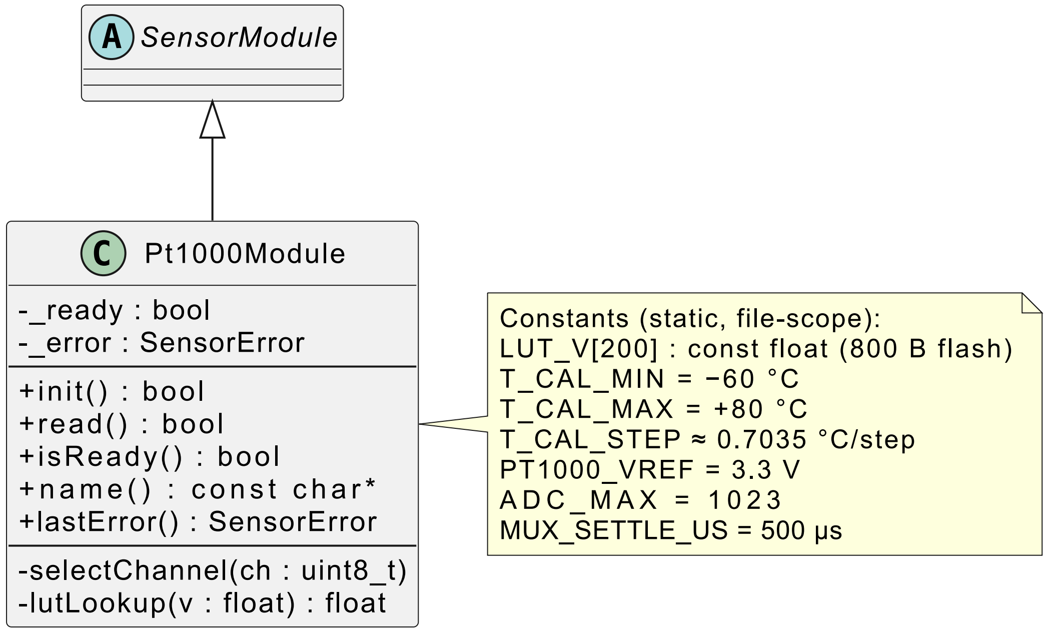
2. Class Diagram



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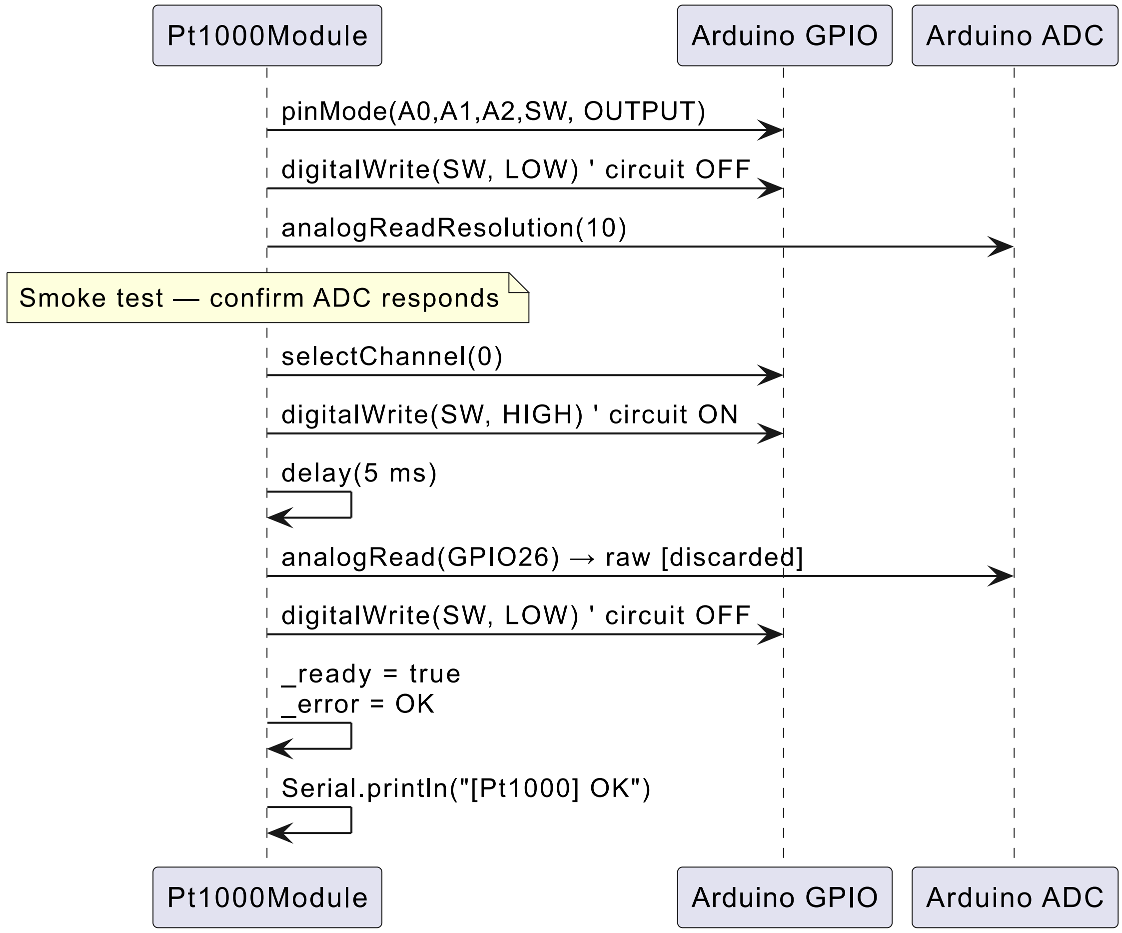
3. Initialization Sequence — `init()`



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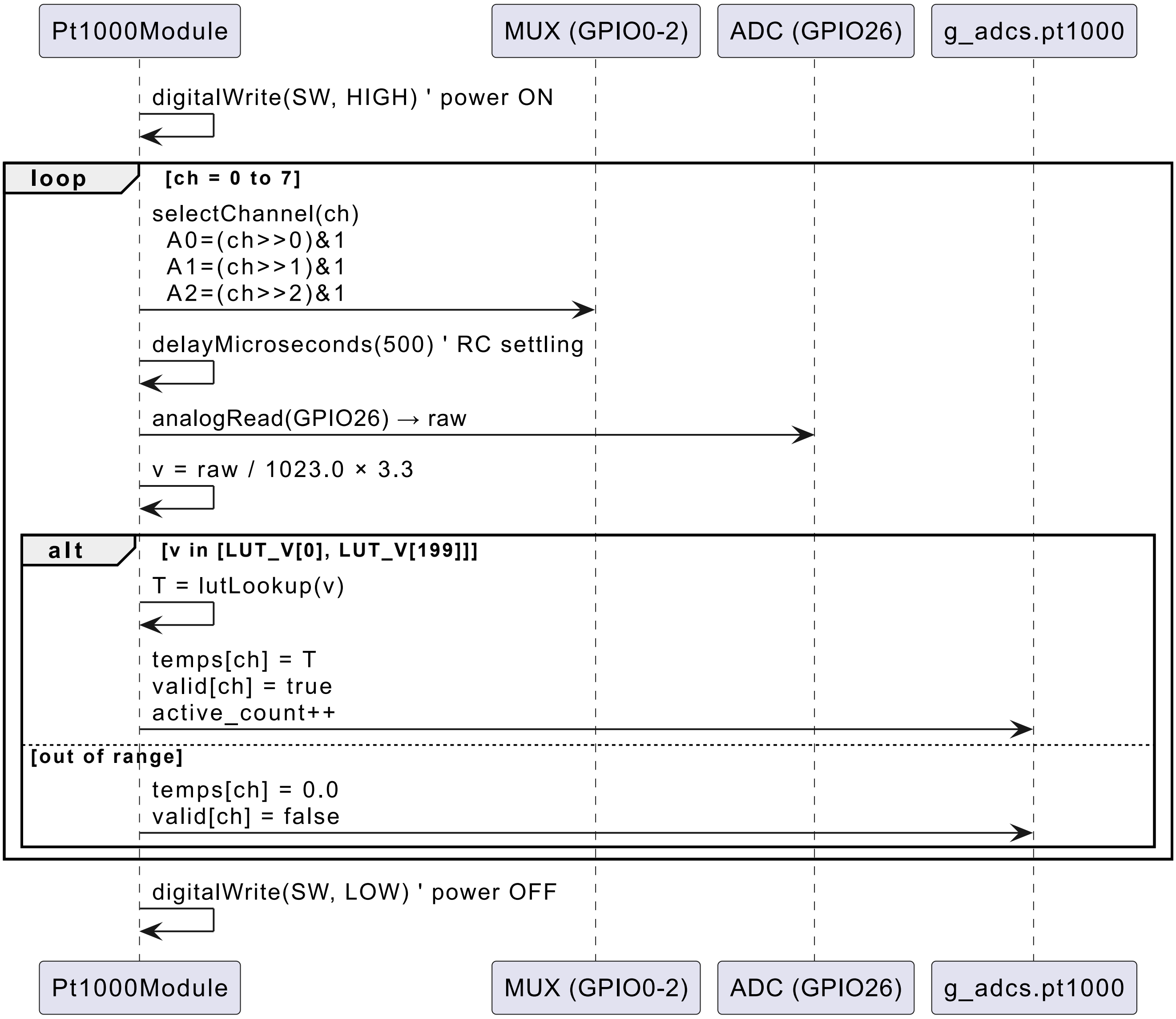
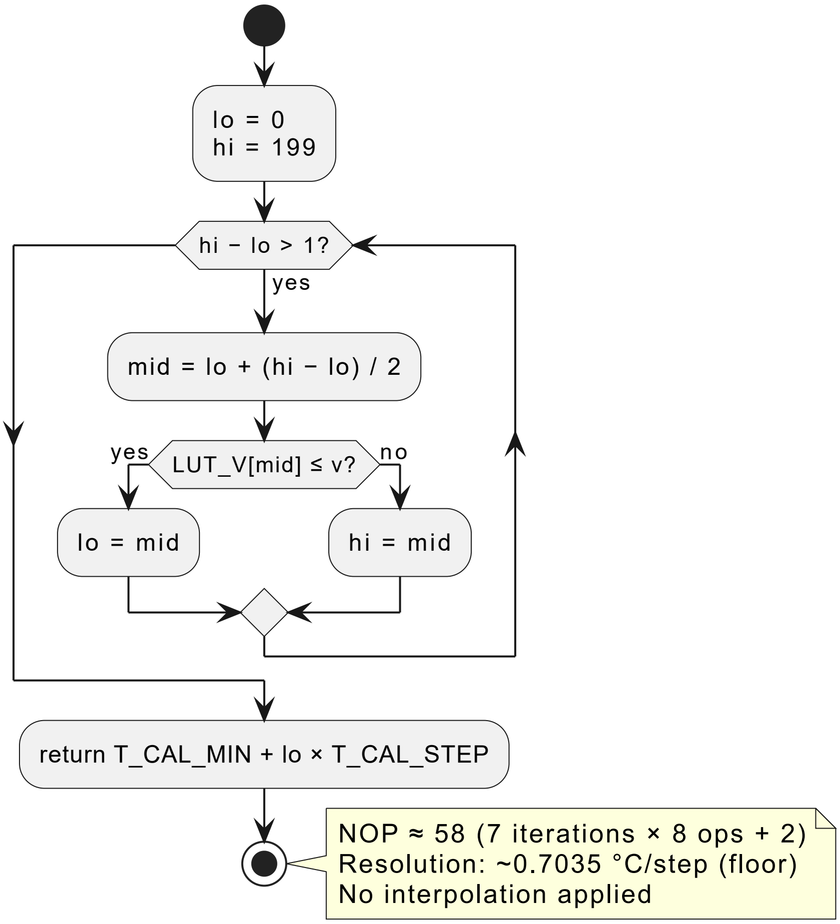
4. Measurement Sequence — `read()` (8 channels)



5. LUT Binary-Search Algorithm — `lutLookup(v)`



Complexity:

Metric	Value
Time	O(log N) — 7 iterations for N = 200
NOP	≈ 58 ops per call
Stack	10 B (lo, hi, mid, v)
Flash	800 B (`LUT_V[200]`)

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6. MUX Channel Selection Truth Table

ch	A2	A1	A0	Pt1000
0	0	0	0	S1
1	0	0	1	S2
2	0	1	0	S3
3	0	1	1	S4
4	1	0	0	S5
5	1	0	1	S6

ch	A2	A1	A0	Pt1000
6	1	1	0	S7
7	1	1	1	S8

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7. Calibration LUT

Source file: `Vout_vs_T_20260603_180610.csv`
200 measured points of steady-state op-amp output voltage at GPIO26 as a function of temperature.

Parameter	Value
Range	−60 °C → +80 °C
Entries	200
Step	≈ 0.7035 °C/entry
V min	−0.0145 V (LUT_V[0])
V max	2.3058 V (LUT_V[199])
Strategy	Floor (no interpolation)